

Test Exam – Regular and Context-Free Languages

ID

Name

Each of the following XXX questions has 4 possible answers of which exactly one is correct. For each question, you may select one or more answers by checking the appropriate boxes. Your test is graded as follows:

- If you select only the correct answer, you receive 2 points.
- If you select 2 answers, one of which is correct, you receive 1 point.
- If you select 3 answers, one of which is correct, you receive 0.41 points.
- If you select no answer or all answers, you receive 0 points.
- If you select only one answer, and it is wrong, you receive -0.67 points.
- If you select 2 answers, that are both wrong, you receive -1 point.
- If you select 3 answers, that are all wrong, you receive -1.25 points.

Note that perfect answers yield a score of XXX and random guessing may give you a negative score. For the logical basis of this score system, see Frandsen and Schwartzbach, *A singular choice for multiple choice*.

Note: this is an open-book exam. You may use books, lecture slides and other notes. The use of electronic devices is *not* permitted.

Question 1

Which statement is wrong?

- ☐ a The class of regular languages is closed under complement
- ☐ b The class of context-free languages is closed under complement
- ☐ c The class of context-free languages is closed under union
- ☐ d The class of regular languages is closed under union

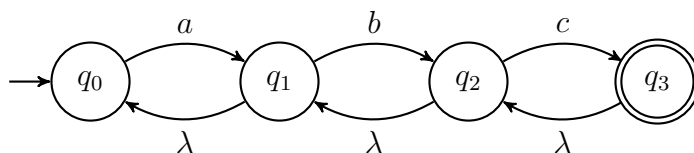
Question 2

Which of the following languages is not regular?

- ☐ a $\{a^m a^m \mid m \geq 0\}$
- ☐ b $\{a^m a^n \mid m \geq n \geq 0\}$
- ☐ c $\{a^m b^m \mid m \geq 0\}$
- ☐ d $\{a^m b^n \mid m \geq 0, n \geq 0\}$

Question 3

Consider the following (non-deterministic) NFA M :



Let N be the minimal, complete, deterministic DFA, such that $\mathcal{L}(M) = \mathcal{L}(N)$. How many states does N have?

- ☐ a 3 states
- ☐ b 4 states
- ☐ c 5 states
- ☐ d 6 states

Question 4

Consider the following context-free grammar:

$$G = \begin{cases} S \rightarrow BS \mid AB \\ A \rightarrow Aaa \mid \lambda \\ B \rightarrow b \end{cases}$$

What is an equivalent *regular* grammar?

a ☐ $\begin{cases} S \rightarrow bS \mid Ab \\ A \rightarrow Aaa \mid \lambda \end{cases}$

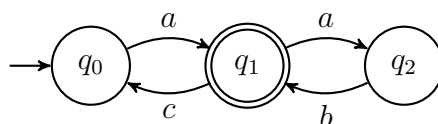
b ☐ $\begin{cases} S \rightarrow bS \mid bA \\ A \rightarrow aA \mid \lambda \end{cases}$

c ☐ $\begin{cases} S \rightarrow bS \mid A \\ A \rightarrow aC \mid b \\ C \rightarrow aA \end{cases}$

d ☐ $\begin{cases} S \rightarrow bS \mid aC \mid b \\ A \rightarrow aC \mid b \\ C \rightarrow aA \end{cases}$

Question 5

Consider the following NFM M :



Which of the following regular expressions does *not* describe the same language as M ?

a ☐ $a(ab + ca)^*$

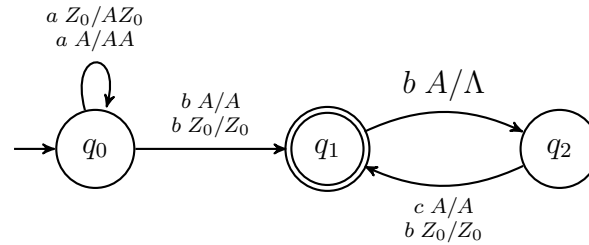
b ☐ $a((ab)^*ca)^*(ab)^*$

c ☐ $a((ca)^*ab)^*(ca)^*$

d ☐ $(ac)^*(ab)^*$

Question 6

Consider the following Push-Down Automaton M .



What is the language accepted by this PDA, $\mathcal{L}(M)$?

- a ☐ $\{a^n b(bc)^m \mid 0 \leq m < n\}$
- b ☐ $\{a^n b(bc)^m \mid 0 \leq m \leq n\}$
- c ☐ $\{a^n b(bc)^m \mid 0 \leq n < m\}$
- d ☐ $\{a^m b(bc)^m \mid 0 \leq m\}$

Question 7

Given are four languages L_1, L_2, L_3, L_4 , with the properties:

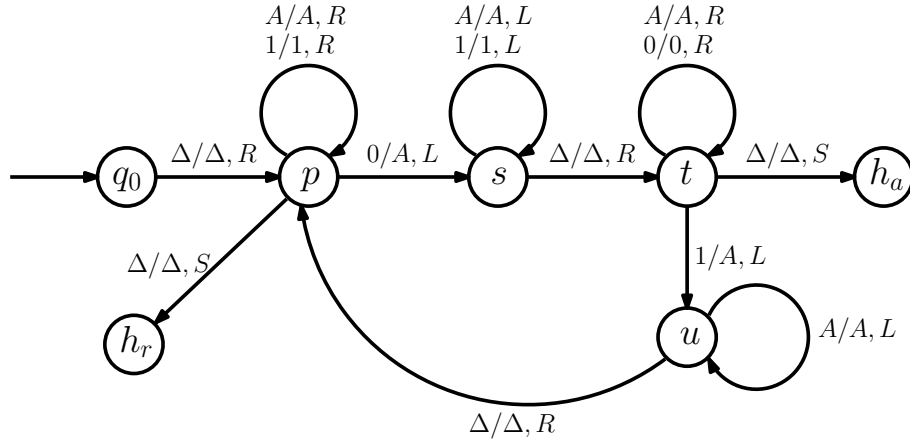
- L_1 is a finite language
- L_2 is a regular language
- L_3 and L_4 are context-free languages

Which of the following languages is *not necessarily* context-free?

- a ☐ $L_1 \cap L_3 \cap L_4$
- b ☐ $L_2 \cap L_3 \cap L_4$
- c ☐ $L_1 \cap L_2$
- d ☐ $L_2 \cap L_3$

Question 1

Consider the following Turing machine T over the alphabet $\Sigma = \{0, 1\}$ with initial state q_0 and tape alphabet $\Gamma = \{0, 1, A\}$:

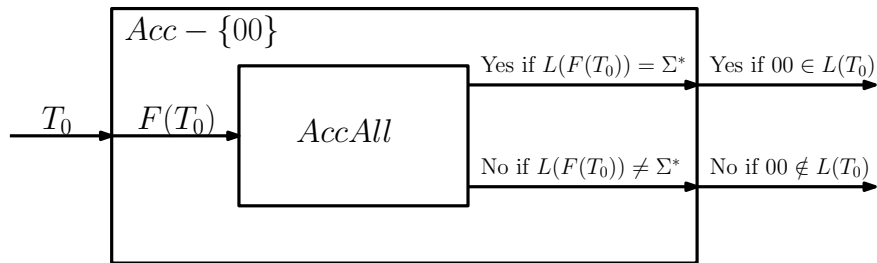


What is the language $L(T)$?

- a ☐ $L(T)$ is the set of all binary strings having an even number of 1's.
- b ☐ $L(T)$ is the set of all binary strings with at most two 1's.
- c ☐ $L(T)$ is the set of all binary strings with more 0's than 1's.
- d ☐ $L(T)$ is the set of all binary strings not containing the substring 010.

Question 2

Let $Acc - \{00\}$ be the following decision problem: Given a Turing machine T_0 , does it accept the string 00? Assume we want to make a reduction from $Acc - \{00\}$ to $AccAll$, where $AccAll$ is the decision problem: Given a Turing machine T_1 , does it accept all strings? Which of the following definitions $F(T_0)$ makes F a correct reduction? (See the following figure for an illustration).



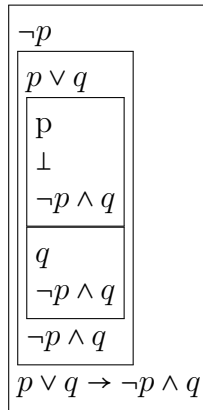
- a ☐ $F(T_0)$ is the Turing machine that starts by replacing the tape contents with the string 00. It then runs T_0 .
- b ☐ $F(T_0)$ is the Turing machine that starts by checking if the input string is 00. If it is, it rewinds the tape head and runs T_0 . Otherwise, it immediately rejects.
- c ☐ $F(T_0)$ is the Turing machine that starts by running the Turing machine T_0 . If this accepts, so does $F(T_0)$ and otherwise it rejects.
- d ☐ $F(T_0)$ is the Turing machine that starts by running the Turing machine T_0 . If this accepts, it checks whether the input string was 00. If so, it accepts. Otherwise it rejects.

Question 3

Consider the following list of proof rules

$(premise), \wedge i, \wedge e_1, \wedge e_2, \vee e, \vee i_1, \vee i_2 \rightarrow i, \rightarrow e, \neg e, \neg i, \perp, \neg\neg e, \forall e, \forall i, \exists e, \exists i, \perp$.

Consider the below proof for $\neg p \vdash p \vee q \rightarrow \neg p \wedge q$, where we have deleted the annotations saying which proof rules have been used.



How many distinct proof rules from the list is being used. If a rule is used several times, then only count it once.

- a ☐ 6
 b ☐ 5
 c ☐ 7
 d ☐ 8

Question 4

Consider the following formula $(p \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow (q \rightarrow r))$. Which of the following statements is true.

- a ☐ The formula is satisfiable but not valid.
 b ☐ The formula is valid but not satisfiable.
 c ☐ The formula is valid and satisfiable.
 d ☐ The formula is neither valid nor satisfiable.

Answers:

First Half

1. B
2. C
3. C (don't forget the trap state)
4. D (actually, not regular according to the book)
5. D
6. B (actually there is a typo: the lowest b must be a c)
7. B

Second Half

1. C (I'm afraid the loop down right is missing 0/0 L)
2. A
3. A
4. A